

SOC ANALYST LEVEL 1

LECTURE 2: NETWORKING FUNDAMENTALS

How a SOC Analyst Reads Communications & Logs

Author: **Nader Anam** | Duration: **3 Training Hours**

LEARNING OBJECTIVES

By the end of this lecture, you will be able to:



Understand **IP Addressing**, Private vs. Public, and NAT.



Master **Ports** & Protocols (TCP/UDP) from a security lens.



Analyze **DNS, HTTP, & HTTPS** logs for anomalies.



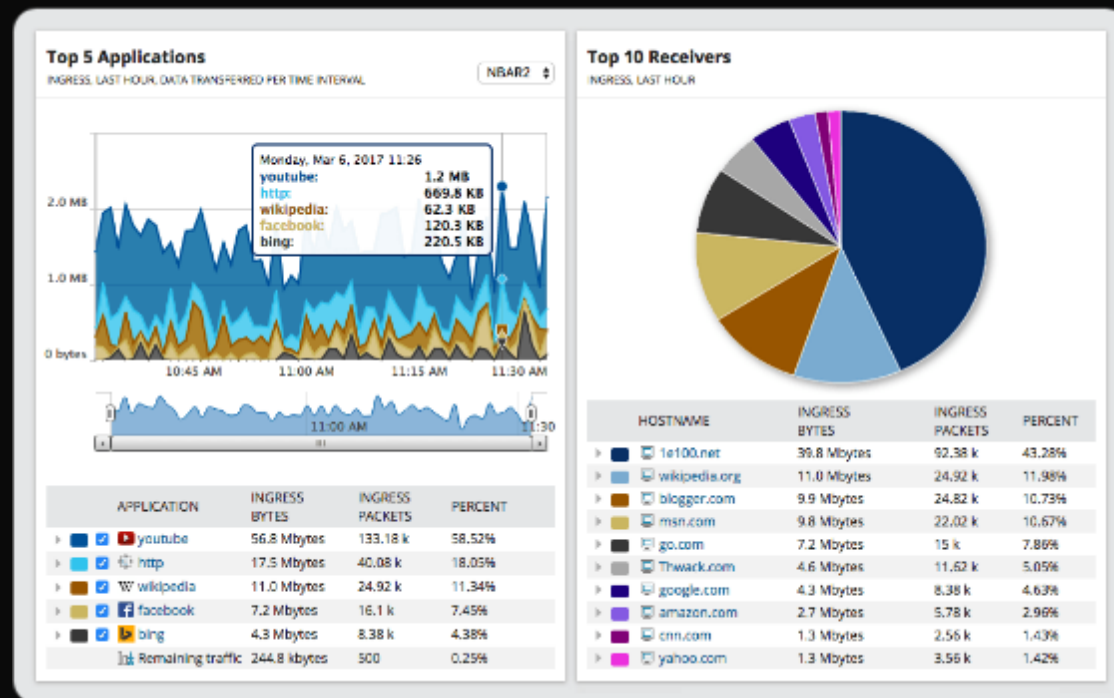
Read **Firewall, Proxy, & VPN** logs to answer: "Who, Where, Why?"

WHY DO SOC ANALYSTS NEED NETWORKING?

A SOC Analyst processes thousands of logs daily. Without networking basics, these are just meaningless strings:

```
10.0.0.15 → 8.8.8.8:53 UDP
192.168.1.20 → example.com:443 TCP
172.16.5.10 → 185.199.108.153:80 TCP
```

You cannot analyze what you cannot read. Your goal is **initial decision-making** based on telemetry.



NOT CCNA: THE SOC VIEWPOINT

We don't study network design or router configuration. We study **behavior**:

WHO CONNECTED?

Source IP identification.

TO WHERE?

Destination IP & Reputation.

WHICH PORT?

Service identification.

WHICH PROTOCOL?

Communication type.

INTERNAL OR EXTERNAL?

Network boundary check.

DATA VOLUME?

Bytes sent vs. received.

HOW SOC SEES THE NETWORK

The network is visible through **Events and Logs**:

- Internal host connecting to external server.
- User visiting a specific URL.
- Host scanning unusual ports.
- DNS query for a suspicious domain.
- Firewall blocking an inbound connection.
- Proxy logging a file download.
- VPN login from a foreign IP.
- Internal host scanning other internal hosts.

Every record can be **an alert** or an **indicator of compromise**.

ELEMENTS OF A NETWORK LOG

Field	Meaning
Source IP	The device that started the connection.
Destination IP	The targeted device or server.
Source Port	Temporary port from the source side.
Destination Port	The service requested (Door).
Protocol	TCP or UDP (mostly).
Action	Allow or Block.
Bytes Sent/Recv	Data volume exiting vs entering.

WHAT IS AN IP ADDRESS?

The address of a device on a network or the internet.

Examples: 192.168.1.15, 8.8.8.8

SOC Insight: IP helps us identify:

- Who started the connection?
- Which device is affected?
- Is the address internal or external?
- Is the IP known for malicious activity?

IP: DEVICE IDENTITY

SOURCE IP VS. DESTINATION IP

SOURCE IP

The address of the device that **initiated** the connection.

DESTINATION IP

The address of the device or server being **connected to**.

10.0.0.15 [SRC] → 8.8.8.8 [DST]

WHY DIRECTION IS CRITICAL

Example 1

10.0.0.15 → 8.8.8.8:53 UDP

Internal host asking an external DNS server. (Normal check needed).

Example 2

8.8.8.8 → 10.0.0.15:53 UDP

External address trying to connect to an internal host on Port 53. (Suspicious!)

Never analyze an IP in isolation. Analyze the direction: Who started it?

PRIVATE VS. PUBLIC IP

PRIVATE IP RANGES

Range	Example
10.0.0.0/8	10.0.0.15
172.16.0.0/12	172.16.5.20
192.168.0.0/16	192.168.1.30

PUBLIC IP

Addresses visible on the global internet.

- 8.8.8.8
- 1.1.1.1
- 185.199.108.153

WHY THIS MATTERS IN SOC?

The analyst needs to know:

- Is the affected device **inside** the organization?
- Did the connection go **out** to the internet?
- Is it an inbound connection from the internet?
- Is an internal host communicating with a suspicious external IP?
- Is there internal communication between hosts suggesting **Lateral Movement**?

```
192.168.1.20 → 45.33.20.10 [Internal to External Traffic]
```



IP = Building Address

Port = Door Number for a specific office.

Port = Door Number for a specific office.

- Port 80: HTTP
- Port 443: HTTPS
- Port 53: DNS

- Port 80: HTTP
- Port 443: HTTPS
- Port 53: DNS

SOURCE VS. DESTINATION PORT

10.0.0.15: 5154 → 8.8.8.8: 5 UDP

4

3

SOURCE PORT (51544)

Usually temporary and random. Low value in initial triage.

DESTINATION PORT (53)

Critical. Tells us the service requested (DNS in this case).

ESSENTIAL PORTS FOR SOC

Port	Service	SOC Relevance
53	DNS	Detecting malicious domains / C2
80 / 443	HTTP/S	Web traffic analysis / Malware downloads
22	SSH	Linux remote access (target for brute force)
3389	RDP	Windows remote access (target for brute force)
445	SMB	Internal file sharing / Lateral Movement
25	SMTP	Email traffic (Phishing indicators)

COMMON PORT != SAFE CONNECTION

Attackers use known ports to **blend in** with normal traffic:

- **Malware** often uses Port 443 to hide C2 traffic within HTTPS.
- **Brute Force** targets Port 22 or 3389.
- **Data Exfiltration** can occur over HTTPS (443).
- **DNS (53)** can be used as a tunnel for stealthy commands.

The port tells you the service, but it doesn't prove the behavior is legitimate.

TCP VS. UDP

TCP

Reliable, connection-oriented. Best for accuracy.

Used by: HTTP/S, SSH, RDP, Email, Databases.

UDP

Fast, lightweight, no-handshake. Best for speed.

Used by: DNS, DHCP, Streaming, NTP.

TCP/UDP IN SOC TRIAGE

Protocol helps you verify the nature of the connection:

```
10.0.0.15 → 8.8.8.8:53 UDP (Likely DNS Query)
```

```
10.0.0.20 → 192.168.1.50:445 TCP (Likely SMB/File sharing)
```

```
192.168.1.30 → 203.0.113.10:3389 TCP (RDP to External IP - Investigation Required)
```

WHAT IS DNS?

Domain Name System: Translates domain names to IP addresses.

User writes:

GOOGLE.COM



DNS returns:

142.250.180.142

From a SOC lens, DNS shows us **where** the hosts are trying to go.

WHY DNS IS CRITICAL IN SOC?

Attacks leave clear footprints in DNS logs:

- Queries for extremely weird domains.
- High frequency of queries to the same domain.
- Connections to newly registered domains.
- Look-alike domains (Typosquatting).
- Malware reaching C2 domains.
- Phishing sites.

Example: `secure-login-update-check.example.net`

DNS LOG ANALYSIS

```
Time: 10:15:22 | Client IP: 10.0.0.25  
Query: secure-login-update.example.net  
Type: A | Response IP: 185.20.44.10 | Action: Allowed
```

Initial Triage:

- Domain name suggests "Login/Update" (Phishing indicator).
- Internal host (10.0.0.25) requested an external IP.
- **Verdict:** Needs Investigation (Check reputation).

HTTP & HTTPS (SOC LENS)

HTTP (PORT 80)

Unencrypted. Reveals URL paths clearly in logs.

HTTPS (PORT 443)

Encrypted. Hides content, but Domain/IP are visible in most logs.

Note: Encryption does NOT equal safety. It only means the connection is private from eavesdroppers.

WHAT TO LOOK FOR IN WEB LOGS?

In Proxy or Web Gateway logs, we analyze:

USER & SOURCE IP

URL & DOMAIN

HTTP METHOD

STATUS CODE

FILE DOWNLOADED

USER-AGENT

Helps detect: Phishing, Malware Downloads, and Data Exfiltration.

PROXY LOG ANALYSIS

```
Time: 14:22:10 | User: sara | Source: 10.0.0.18  
URL: hxxps://invoice-review.example-download.com/file.exe  
Action: Allowed | File: .exe | Size: 4.8 MB
```

Analysis:

- User "sara" downloaded an executable (.exe).
- Domain name suggests "Invoice review" (Common lure).
- **Verdict:** Suspicious Download - Investigate the file & URL.

WHAT IS A FIREWALL LOG?

Records allowed or blocked connections at the network boundary.

Field	Meaning
Source / Dest IP	The origin and target addresses.
Dest Port	The service targeted.
Action	Allow or Block.
Rule Name	Which policy triggered this action.
Bytes	Data volume.

FIREWALL LOG: NORMAL CASE

```
Time: 09:10:45 | SRC: 10.0.0.22 | DST: 185.199.108.153  
Port: 443 | Protocol: TCP | Action: Allow | Bytes: 1200
```

Analysis:

- Internal host connecting to external HTTPS via Port 443.
- Data volume is small. Connection is Allowed.
- **Verdict:** Normal activity (unless linked to other alerts).

FIREWALL LOG: SUSPICIOUS CASE

```
Time: 01:33:12 | SRC: 10.0.0.45 | DST: 45.77.20.10  
Port: 4444 | Protocol: TCP | Action: Allow | Bytes Out: 90 MB
```

Analysis:

- Port 4444 is uncommon. Data volume (90MB Out) is high.
- Time is late at night.
- **Verdict:** High Suspicion - Possible Data Leak or Shell connection.

WHAT IS NAT?

Network Address Translation:

Many internal devices (Private IPs) share one single **Public IP** when communicating with the internet.

`[10.0.0.15, 10.0.0.16] → [NAT GATEWAY] → 203.0.113.5 (Public)`

NAT & THE SOC ANALYST

When an external report says "203.0.113.5 attacked us," it represents your **entire gateway**.

To find the real attacker, the analyst must ask:

- Who was the actual internal device?
- Do we have **NAT logs** to map the Public IP back to the Private IP?
- What was the exact timestamp and Source Port?

WHAT IS A VPN LOG?

Records remote users logging into the corporate network.

FIELDS

Username, Source Public IP, Country, MFA Status,
Device Info, Session Duration.

ANALYTIC USES

Impossible Travel, Brute Force, Compromised
Credentials.

VPN LOG ANALYSIS

```
User: ahmed | Source IP: 185.20.40.9 | Country: Unknown  
Login: Success | MFA: Not Used | Time: 02:10 AM
```

Analysis:

- Successful login from a foreign IP at 2:00 AM.
- **MFA was NOT used.**
- **Verdict:** Potential Account Compromise - Investigate activity.

WHAT IS AN IDS/IPS LOG?

IDS (DETECTION)

Sends alerts for suspicious patterns but doesn't stop them.

IPS (PREVENTION)

Detects and **blocks** the connection automatically.

Alerts include: Port Scans, SQL Injection, Exploits, and C2 traffic.

IDS ALERT: PORT SCAN

Alert: Possible Port Scan | Source IP: 10.0.0.55
Destination Range: 10.0.0.1 - 10.0.0.254 | Action: Detected

Analysis:

- Internal host scanning many other hosts.
- Could be an admin tool OR a compromised device doing discovery.
- **Verdict:** Needs Investigation. Check device owner/role.

COMMAND & CONTROL (C2)

The communication channel used by malware to talk to the attacker.

IoCs (Indicators):

- Repetitive small connections (Beaconing).
- Random or long domain names.
- Uncommon ports.



BEACONING PATTERN

Connection every 60 seconds exactly.

SUSPICIOUS NETWORK PATTERNS

- Connections every minute to external IP.
- Huge data exfiltration outside working hours.
- User host connecting to external RDP.
- DNS queries for long, random domains.
- Internal Port Scanning.
- Exe/Zip downloads from unknown domains.
- Connections to countries with no business ties.
- Encrypted PowerShell connections.

DON'T RUSH TO JUDGMENT

Not everything "weird" is an attack. Examples of normal activity:

- Software updates.
- Authorized admin tools.
- Cloud service background traffic.
- IT Team security scans (Planned).
- Backups occurring late at night.

turns a log into

INVESTIGATION METHODOLOGY

Ask these questions for **every** network log:

- Who is the Source?
- Where is the Destination?
- Internal or External?
- What is the Port/Protocol?
- Allowed or Blocked?
- What was the volume?
- Is the target reputable?
- Is this normal for this user?
- **Final Decision: Close or Escalate?**

INVESTIGATION TEMPLATE

Time: [ISO Format]
Source IP / Host / User: [...]
Destination IP / Domain: [...]
Port / Protocol: [...]
Action: [Allow/Block]
Observed Activity: [Narrative]
Why Suspicious: [Rationale]
Severity: [L/M/H/C]
Decision: [Close/Escalate]

PRACTICAL DRILL 1

Analyze this log:

```
Time: 10:00:15 | SRC: 10.0.0.15 | DST: 8.8.8.8  
Port: 53 | Protocol: UDP | Action: Allow
```

Questions:

- Service?
- Direction?
- Normal or Suspicious?
- Should we escalate?

DRILL 1: SOLUTION

- **Service:** DNS.
- **Direction:** Internal to External.
- **Analysis:** Normal activity (Google DNS) unless the queries contain suspicious domains.
- **Verdict:** Close (No escalation needed alone).

PRACTICAL DRILL 2

Analyze this log:

```
Time: 23:45:01 | SRC: 10.0.0.33 | DST: 91.200.10.5  
Port: 443 | Protocol: TCP | Action: Allow  
Frequency: Every 60 seconds
```

Questions:

- Is Port 443 always safe?
- What is suspicious here?
- Decision?

DRILL 2: SOLUTION

- **Suspicion:** Connection every 60s suggests **Beaconing**.
- **Verdict:** Needs Investigation (Check IP reputation).
- **Evidence needed:** Process name, DNS query, EDR alerts.

PRACTICAL DRILL 3

Analyze this log:

```
Time: 02:05:44 | SRC: 10.0.0.77 | DST: 203.0.113.90  
Port: 3389 | Action: Allow | User: unknown
```

Questions:

- Service? Source of danger?
- Escalate or Close?

DRILL 3: SOLUTION

- **Service:** RDP.
- **Danger:** Internal host connecting to **External RDP** late at night.
- **Verdict:** High Suspicion - Escalate to Tier 2.

PRACTICAL DRILL 4

Analyze this alert:

```
Alert: Internal Port Scan | SRC: 10.0.0.50  
DST: 10.0.0.1 - 10.0.0.254 | Ports: 22, 80, 443, 445, 3389
```

Is this always a hack? What evidence do you need?

DRILL 4: SOLUTION

- **Analysis:** Could be an authorized security scan.
- **Required Evidence:** Asset owner, planned activity schedule, user context.
- **Verdict:** Investigation Required. Do not close immediately.

WORKSHOP BREAKOUT

Divide into groups. Analyze one log per group:

GROUP 1

Suspicious DNS Query.

GROUP 2

Exe download from unknown site.

GROUP 3

External RDP connection.

GROUP 4

Internal Port Scan.

GROUP 5

VPN Login from new country.

FULL TRAINING SCENARIO

Alert: "HR-LAPTOP-12" ran PowerShell late at night.

```
1. 10.0.0.25 → secure-update-login.example.net (DNS Query)
2. 10.0.0.25 → 185.20.44.10:443 every 60 seconds
3. 10.0.0.25 → 45.77.20.10:4444 Allowed
```

Identify IoCs and decide: Escalate or Close?

SCENARIO ANALYSIS

- **IoCs:** Late-night PowerShell, Beaconing (60s), Port 4444.
- **Most Dangerous:** Port 4444 (Possible Shell).
- **Verdict:** Realized Compromise - **Escalate to Tier 2.**

TOP 10 ROOKIE MISTAKES

- Thinking every external IP is a hack.
- Thinking Port 443 is always safe.
- Ignoring Source vs. Dest direction.
- Forgetting NAT exists.
- Ignoring timing and frequency.
- Ignoring data volume.
- Judging DNS based on text alone.
- Not correlating with EDR alerts.
- Using "Confirmed Breach" without proof.
- Analyzing logs in isolation.

GOLDEN RULES OF PHRASING

INCORRECT PHRASING

"This is a hack!"

"This log is garbage."

PROFESSIONAL PHRASING

"Anomalous behavior detected in vectors A & B."

"Closing as False Positive after verifying records X & Y."

LECTURE SUMMARY

- IPs (Private vs Public) identify the "Who."
- Ports identify the "Service."
- DNS identifies "Intent."
- Firewalls/Proxies control the "Flow."
- Analysis methodology must be evidence-driven.

CLASSROOM REVIEW

Quick Review (Answers in mind):

1. Define SRC vs DST IP.
2. Why is Destination Port critical?
3. Which service uses Port 53?
4. Is Port 443 always safe?
5. Why is NAT important for analysts?
6. What is Beaconing?
7. Name 3 IoCs in network traffic.
8. How does DNS reveal C2?
9. What's the difference between IDS & IPS?
10. When should you escalate an RDP connection?

LIVE POP QUIZ

Q1: 192.168.1.20 is a:

A) Public IP | B) Private IP

Q2: Port 53 is usually:

A) DNS | B) HTTP

Q3: Connection every 60s to external IP is:

A) Printing | B) Beaconing

QUIZ ANSWER KEY

Q1: **B** (Private IP - Class C range).

Q2: **A** (DNS service).

Q3: **B** (Beaconing - typical malware behavior).

HOMEWORK ASSIGNMENT

Analyze 4 logs (details in portal) using the **SOC Investigation Template**.

Include: SRC, DST, Port, Protocol, Activity, Verdict, and Severity.

COMING UP NEXT

LECTURE 3: WINDOWS SECURITY

Users, Processes, Registry, and the Power of Event Logs

IMAGE SOURCES



https://cyberone.security/hubfs/Imported_Blog_Media/SolarWinds-nta-network-traffic-analysis.png

Source: cyberone.security



https://s3.ap-south-1.amazonaws.com/webasha-blog/uploads/images/202507/image_750x_687b455e6df16.webp

Source: www.webasha.com